

Name: A ProletariatDate: 03/10/2026Class: Period 6 Engineering Systems

MyDesign Portfolio

Element C Initial Template

Element C: Presentation and Justification of Solution Design Requirements

Design Requirement #1 (High Priority)

Prevent the chair from rolling under the aux cable box and prevent it from getting wedged behind it. Prior solutions have done this by creating a cover that attaches directly on to the wall, creating a flush seal. To be measured, if the chair can fit behind the audio box at any sitting angle [60°–90°], we have **failed**.

Design Requirement #2 (High Priority)

Protecting the cable from impacts with an isolated system or attachment; making it more durable. Previous solutions have done this with braided cables or protective coverings. If the cable can still be broken with an impact when the design solution is present, we have **failed**

Design Requirement #3 (High Priority)

Our third priority is to create a design that is secure enough while also offering easy access to all the cables. Previous solutions have done this with a hinge, or an easily removable cover. If cables are locked behind the solution, we have **failed**.

Design Requirement #4 (High Priority)

Our 4th highest priority is to make our design "invisible". Prior solutions have been bright colors and unsightly, the antithesis of our requirement. Our stakeholder was adamant that she didn't want to notice the design solution. If the design solution is immediately visible and harms the aesthetic of the desk area, we have **failed**.

Design Requirement #5 (High Priority)

Our 5th priority is to create and maintain a cable organization system that can easily keep all of the cables in their place. Prior solutions are colorful, and our stakeholder's current set-up doesn't hide any cables, making the area a bit of an eyesore. Our stakeholder has a lot of cables that she doesn't want to see, which goes hand-in-hand with her desire to make a design blend in. If the cables get tangled or clutter the area, we have **failed**.

Design Requirement #6 (Low Priority/Stretch Goals)

Our 1st stretch goal is to lower the existing mounted position of the aux cable box. Redrilling the existing mounting position would prevent the chair from getting below the box, and thus preventing the wires from getting broken. However, this would make it harder to access the wires since our stakeholder is tall and wouldn't take kindly to having to bend down more to access the cables. If the cables can still get broken even with a lower height, we have **failed**.

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Design Requirement #7 (Low Priority/Stretch Goals)

Our 2nd stretch goal is to create a new mounting system for the box itself, as the current one is ineffective. Other rooms in the school, and prior solutions, have screwed the box into the wall itself. However, we would need explicit permission from the school to achieve this. If the box still has a chance to fall down, we have **failed**.

Our contact with our stakeholder:



Reynie Benisch (Student) <1645855@fcpsschools.net>
to Kate, me, Johannes, David ▾

Wed, Mar 18, 12:01 PM ☆ 😊 ↩ ⋮

Hi Ms. Anderson, hope your week is going well.

Our team has spent the last week making a list of requirements to solve the problem you have with the aux cables breaking, and now we would like your feedback:

Design Requirement #1 (High Priority)

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Does this list meet your requirements for this project? Are there any other requirements you would like us to add, or any of these requirements that you wouldn't like us to prioritize? Any feedback you have would be appreciated.

Thanks and best,
Reynie Benisch, et. al.



Reynie Benisch (Student) <1645855@fcpsschools.net>
to Kate, Kate, me, Johannes, David ▾

Mon, Mar 23, 12:00 PM ☆ 😊 ↩ ⋮

Hi Ms. Anderson,

Just following up about giving feedback on our design requirements that we shared. We still would appreciate all feedback as we move forward on solutions to your particular problem. Here's our preliminary list of requirements once again:

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Thank you for your time,
Reynie Benisch, et. al.



Anderson, Kate <kganderson1@fcps.edu>
to Reynie, Kate, me, Johannes, David ▾

Mar 23, 2026, 7:31 PM ☆ 😊 ↩ ⋮

This list of requirements looks great. I see no changes that need to be made.

Kate Anderson

Kate Anderson (she/her)

Marshall High School
Earth Science 1
IB Environmental Systems and Societies

From: Reynie Benisch (Student) <1645855@fcpschools.net>

Sent: Monday, March 23, 2026 11:59 AM

To: Anderson, Kate <kganderson1@fcps.edu>; Kate Anderson <kganderson1@fcpschools.net>

Cc: Emerson Stering (Student) <1911578@fcpschools.net>; Johannes Jangam (Student) <1709107@fcpschools.net>; David Tsvetovat (Student) <1584127@fcpschools.net>